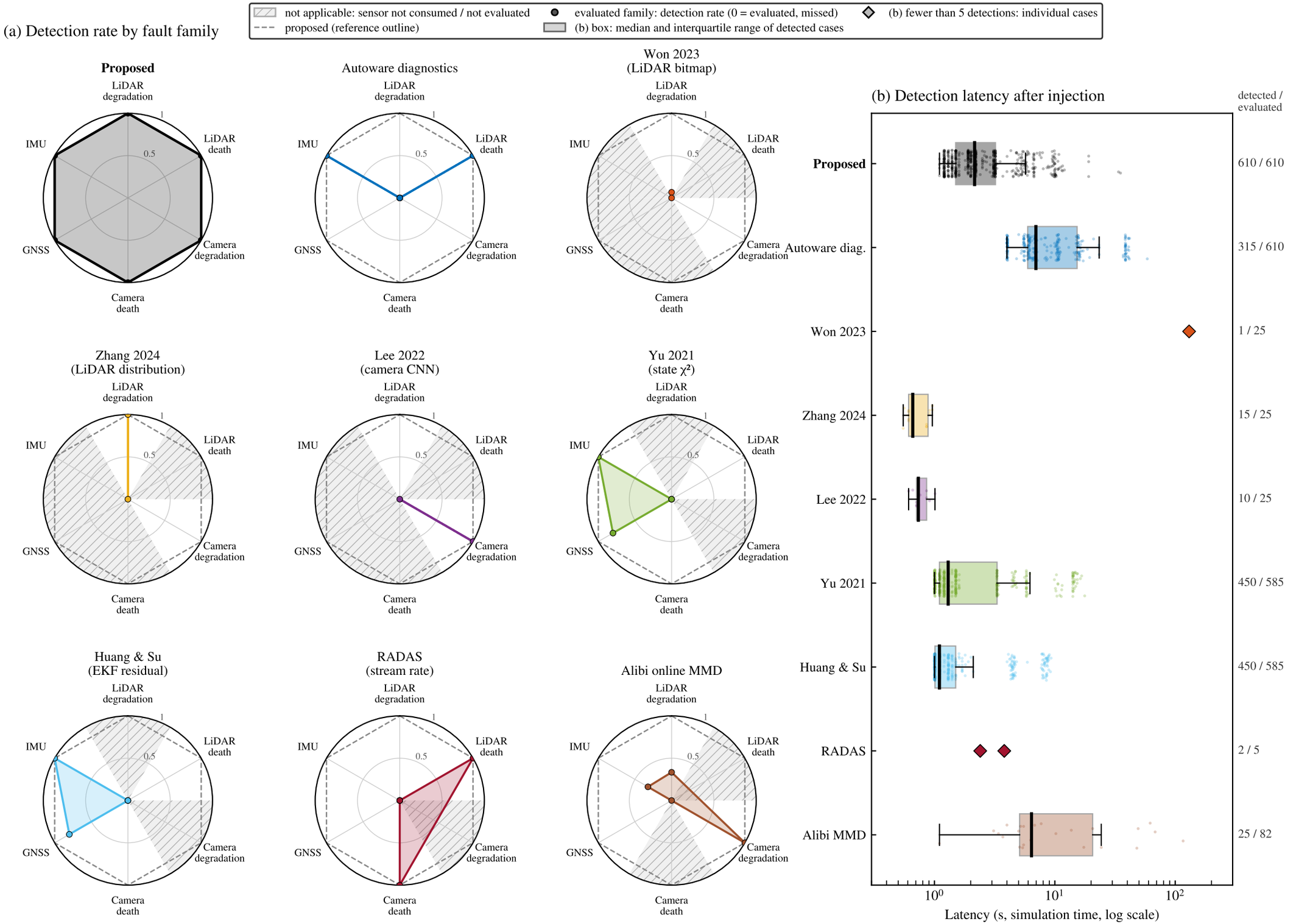

(a) Detection rate by fault family
not applicable: sensor not consumed / not evaluated
proposed (reference outline)
evaluated family: detection rate (0 = evaluated, missed)
(b) box: median and interquartile range of detected cases
(b) fewer than 5 detections: individual cases
Proposed
Autoware diagnostics
Won 2023 (LiDAR bitmap)
Zhang 2024 (LiDAR distribution)
Lee 2022 (camera CNN)
Yu 2021 (state χ²)
Huang & Su (EKF residual)
RADAS (stream rate)
Alibi online MMD
LiDAR degradation
LiDAR death
Camera degradation
Camera death
IMU
GNSS
0.5
1
(b) Detection latency after injection
detected / evaluated
Proposed
Autoware diag.
Won 2023
Zhang 2024
Lee 2022
Yu 2021
Huang & Su
RADAS
Alibi MMD
610 / 610
315 / 610
1 / 25
15 / 25
10 / 25
450 / 585
450 / 585
2 / 5
25 / 82
10⁰
10¹
10²
Latency (s, simulation time, log scale)